AAE338 THERMAL SCIENCES

(3 credits)

COURSE INFORMATION/DESCRIPTION:

A foundational course encompassing a broad spectrum of subjects drawn from the fields of engineering thermodynamics, fluid mechanics, heat transfer, and combustion. This course serves as an introductory level for students, equipping them with the essential knowledge needed to engage in advanced thermal analyses of propulsion and energy systems.

LEARNING RESOURCES AND REFERNCE TEXT

PURDUE BRIGHTSPACE is used to organize the course materials such as announcements, course notes, homework assignments, grades, and solutions. It is important to regularly check Brightspace for the latest update. Lecture notes will be provided before each lecture in Brightspace.

REQUIRED TEXTBOOK McGraw-Hill Create™ Book, Thermal Sciences (AAE 338), Print version ISBN: 9798219038389; E-Book version ISBN: 9798219038372

ASSIGNMENTS:

Homework

Weekly homework is assigned. All homework assignments are intended to be completed independently. While seeking assistance from the instructor, TA, or fellow students is encouraged when necessary, the work submitted should reflect one's individual effort.

Quizzes

This course comprises two categories of quizzes: pre-work quizzes and in-lecture quizzes. Pre-work quizzes are associated with the assigned readings prior to lectures, and these quizzes are accessible before the lecture starts. During lectures, in-lecture quizzes will be administered. These quizzes are intended to reinforce comprehension, foster class participation, and serve as both a teaching and learning evaluation tool.

Project

There is a course project assignment to be completed throughout the semester. A project report is expected to be submitted toward the conclusion of the term.

Exams

There are two exams for the course that are planned as follows: Mid-Exam near the middle of the semester and Final exam at the end of the semester.

COURSE LEARNING OBJECTIVES:

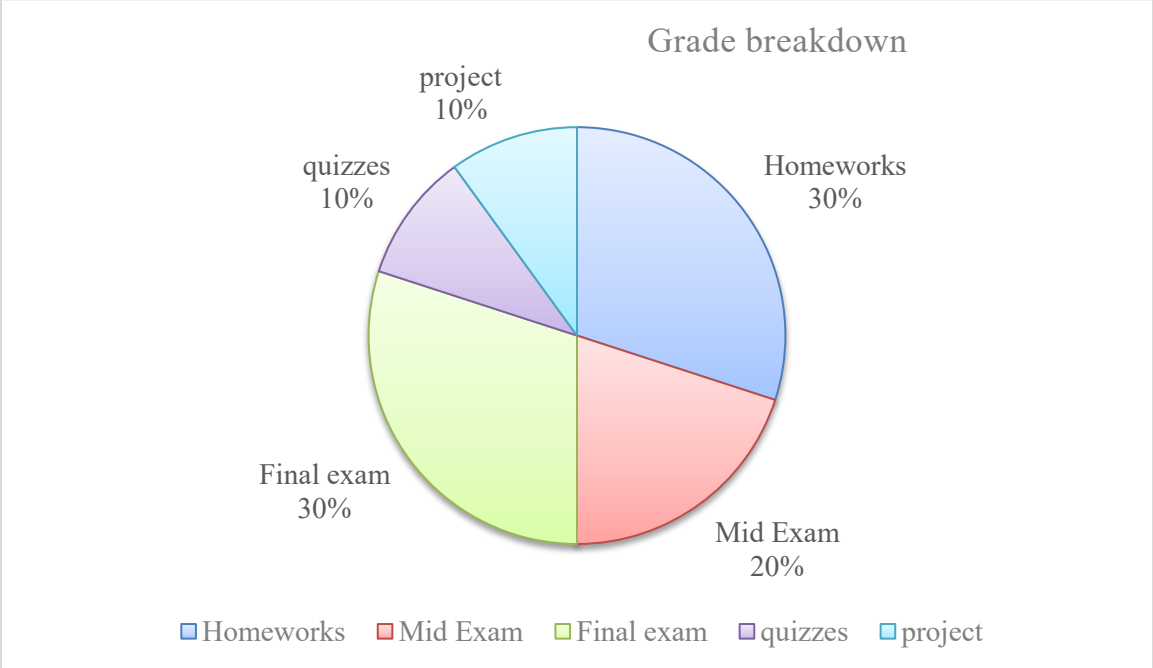
Learning Outcomes	Learning Objectives (LO)
Learning Outcome 1: Reinforce understanding of thermo-fluid concepts and analysis relevant to propulsion applications	LO 1.1: Identify the types of control volume analysis and perform the analysis for each type
	LO 1.2: Develop a deeper understanding of the second law of thermodynamics and entropy and apply them to various thermodynamic problems
	LO 1.3: Perform analysis for one-dimensional compressible flows under the effects of area change, shocks, friction (Fanno flow), and heat

	transfer (Rayleigh flow)
Learning Outcome 2: Gain comprehensive knowledge of heat transfer and perform heat transfer analysis for engineering problems	LO 2.1: Describe different modes of heat transfer and the physical mechanism of each mode
	LO 2.2: Apply concepts and analysis to steady and transient heat conduction problems
	LO 2.3: Develop analysis for convection heat transfer problems involved in internal and external flows
	LO 2.4: Provide a scientific explanation to the green-house effect and apply concepts and analysis to radiation heat transfer problems
	LO 2.5: Integrate analysis of different modes of heat transfer to problems involving multi-mode simultaneous heat transfer
Learning Outcome 3: Acquire basics of combustion and perform calculations of simple combustion systems	LO 3.1: Describe gas mixture and chemical reactions and evaluate mixture properties
	LO 3.2: Determine adiabatic flame temperature of simple combustion systems
	LO 3.3: Relate chemical equilibrium to the second law of thermodynamics and develop a criterion for chemical equilibrium
	LO 3.4: Determine chemical equilibrium state of simple combustion systems

EVALUATION:

Course Grade

The final grades are computed utilizing the following pie chart as an approximate guide.



Grade Scale

The course uses the course average as the reference scale for assigning the final grades. *The average among different semesters does not vary significantly. However, if the average during a semester is higher, this scheme will be adjusted accordingly.*

Grade	Justification
A, A±	Significantly above average*
B, B±	Near and above average
C, C±	Slightly below average
D, D±	Significantly Below average
F	Extremely poor performance